ADVERSARIAL PC REVIEW · v6.45 · STOC/FOCS/ICALP STANDARD

v6.42-v6.44 Closure Theorem: Program Committee Review

Simulated review by an adversarial PC consisting of experts in computational complexity, information theory, communication complexity, learning theory, proof complexity, distributed computing, quantum computing, and TCS at large. Every claim independently verified against the strongest existing frameworks. Every [PROVEN] tag ignored. Maximum skepticism applied per reviewing mandate.

FINAL VERDICT	C. Publishable synthesis, no new theorem

SUBSUMING FRAMEWORKS: Standard complexity theory (Arora-Barak 2009) + 10 other published frameworks

NOVELTY: 0-2/10 across 20 claims; average 1.0/10

VENUES: REJECTED at STOC, FOCS, CCC, ICALP, COLT, NeurIPS Theory, JACM, SICOMP

Executive Summary

The submission chain (v6.42 → v6.43 → v6.44) is treated as a single evolving paper. Every claim is independently verified against the TCS literature. All internal [PROVEN] tags are ignored.

Findings: The v6.42 main theorem is **falsified** by P/poly advice (standard material). The v6.42 Lemmas 2-10 are **already known** (Brent 1974, Mehlhorn-Schmidt 1982, PCP 1998, Denning 1968, Buhrman-de Wolf 2002). Lemma 11 is **unsupported**. Lemma 12 (mutual irreducibility) is **incorrect** — contradicted by complexity-class inclusions. All 7 v6.43 counterexamples are **standard TCS results**. The v6.44 7-primitive closure is **HYPOTHESIZED and definition-dependent**, conditional on the open conjecture BQP $\not\subset$ P/poly.

Verdict: C. The submission contains a useful synthesis (organizing known complexity-theory results under a non-standard 5/7-primitive taxonomy) but no new theorem. Every claim is subsumed by 11 published frameworks.

Part 1 — Independent Verification of Every Claim

1.1 v6.42 Main Theorem (Closure Claim)

Claim: Under A1-A5, every practical computational advantage reduces to one of {C, A, S, V, E}.

Independent verification: A1 (Church-Turing) is a physical claim, not a theorem. A4 (10 axes exhaustive) is **false** — randomness (BPP), advice (P/poly), proof length (PCP), interaction rounds (IP), entanglement (BQP), asynchrony (FLP), average-case (DistP) are all standard axes not in v6.42's list. "Exactly five primitives" is **definition-dependent**. "No sixth primitive exists" is a universal negative over an undefined set, mathematically vacuous.

Falsified by $L = \{1^n : \text{n-th TM halts}\}$ — undecidable but $\in$ P/poly with 1-bit advice. (Standard material: Arora-Barak exercise 6.5.)

Verdict: INCORRECT as stated. Falsified by P/poly.

1.2 v6.42 Lemma 1 (Axis Exhaustiveness)

Claim: 10 axes {S, W, T, D, C, V, Q, P, L, A} are exhaustive for TMs.

Independent verification: Argument given ("TM = tape + head + state + transition") proves only 4 mechanical components, not 10 complexity axes. Missing standard axes: randomness (Sipser ch. 10), advice (Karp-Lipton 1980), proof length (Cook-Reckhow 1979), interaction rounds (Babai-Moran 1988), entanglement (Vidal 2000), asynchrony (FLP 1985).

Verdict: INCORRECT. At least 6 standard axes missing.

1.3 v6.42 Lemmas 2-11 (Per-Axis Bounds)

Lemma	Claim	Independent verdict
L2	S(r) ⩾ K(f) — Kolmogorov bound	ALREADY KNOWN (Li-Vitányi 2008). K(f) uncomputable.
L3	W(r) ⩾ K(f)	STRICT COROLLARY of L2
L4	T(r,p) ⩾ ⌧W/p⌧ — Brent	ALREADY KNOWN (Brent 1974)
L5	D(r) ⩾ ⌧log⌧ W⌧	ALREADY KNOWN (Brent 1974)
L6	C(f) ⩾ rank(M_f)	ALREADY KNOWN (Mehlhorn-Schmidt 1982)
L7	V ⩾ ⌧log(1/ε)⌧	ALREADY KNOWN (PCP, Arora et al. 1998)
L8	Q(f) ⩾ log⌧|outputs|	ALREADY KNOWN (Buhrman-de Wolf 2002)
L9	P ⩽ ⌧W/D⌧	STRICT COROLLARY of Brent
L10	L ⩾ working-set size	ALREADY KNOWN (Denning 1968)
L11	A ⩾ K(distribution)/n	UNSUPPORTED — no citation; appears invented

Verdict: 8 already known, 1 strict corollary, 1 unsupported.

1.4 v6.42 Lemma 12 (Mutual Irreducibility)

Claim: 5 primitives are pairwise mutually irreducible.

Independent verification: Reducibility is informal ("algorithm using only Π" is undefined). Separation problems are sketches, not formal. **Contradicted** by v6.44's own Agent F1: under

formal complexity-class definitions (E=P, C=L, S=NC, A=Θ⊠^P, V=NP), the 5 form a partial order (L, NC ⊆ P ⊆ NP; P ⊆ Θ⊠^P) with 9 inclusions proven, 0 separations proven.

Verdict: INCORRECT. Contradicted by complexity-class inclusions.

1.5 v6.44 7-Primitive Closure

Claim: Under A1+A5, every practical advantage reduces to one of {C, A, S, V, E, Adv, Q}.

Independent verification: 7-primitive set is non-standard. Closure claim is unproven — argued by elimination from 10 candidates, not positive proof. Known gaps:

(a) **Quantum advice (BQP/qpoly)**: may be strictly stronger than BQP ∪ P/poly (Aaronson 2010).

(b) **Average-case / smoothed** (Spielman-Teng 2001): not included as primitive.

(c) **Distributed asynchrony** (FLP 1985): not included as primitive.

(d) **Post-quantum theories**: excluded by A1 (which is a thesis, not theorem).

Verdict: UNSUPPORTED. HYPOTHESIZED, definition-dependent, conditional on BQP ⊄ P/poly (open).

Part 2 — Falsification of the 4 Critical Claims

2.1 "The 10 axes are exhaustive" — FALSIFIED

P/poly (Karp-Lipton 1980) = advice axis. BPP (Gill 1977) = randomness axis. BQP (Bernstein-Vazirani 1997) = entanglement axis. None in v6.42's 10. **Falsified.**

2.2 "Exactly five primitives" — DEFINITION-DEPENDENT

5 primitives are not standard complexity objects. Under v6.44's F1 formalization, they map to {L, NC, P, NP, Θ⊠^P} — 5 of dozens of standard classes. Choice is conventional, not derived. **Definition-dependent, not mathematically proven.**

2.3 "Every practical advantage" — VAGUE AND CIRCULAR

"Practical" excludes constant-factor advantages (most engineering wins). "Polynomial reduction" presupposes the equivalence being defined. "Asymptotic" presupposes input size notion (unclear for distributed protocols). **Vague and circular.**

2.4 "No sixth primitive exists" — FALSIFIED (5-prim); UNSUPPORTED (7-prim)

5-primitive version: **falsified** by P/poly advice (v6.43 Counterexample A). 7-primitive version: **HYPOTHESIZED, not proven**. v6.44's candidate search (10 candidates) is not a proof of nonexistence. BQP/qpoly remains an open 8th-primitive candidate.

Part 3 — Novelty Matrix

For each major claim: closest known theorem, difference, reduction, novelty score (0 = already published, 10 = genuinely new).

#	Claim	Closest known	Novelty
1	5 primitives exhaustive	Cook's TM simulation theorem	0/10
2	10 axes exhaustive	Sipser ch. 7-10 taxonomy	0/10
3	L2: S(r) ⩾ K(f)	Li-Vitányi 2008 ch. 2	0/10
4	L4: Brent's theorem	Brent 1974	0/10
5	L6: rank(M_f) bound	Mehlhorn-Schmidt 1982	0/10
6	L7: PCP amplification	Arora et al. 1998	0/10
7	L8: query lower bound	Buhrman-de Wolf 2002	0/10
8	L10: working-set bound	Denning 1968	0/10
9	L11: amortization ⩾ K/n	No known source	1/10
10	L12: 5×5 mutual irreducibility	None — contradicted	0/10
11	CE A: P/poly advice	Arora-Barak ex. 6.5	0/10
12	CE D: IP=PSPACE	Shamir 1992 FOCS	0/10
13	CE E: FLP	Fischer-Lynch-Paterson 1985	0/10
14	CE G: average-case	Spielman-Teng 2001	0/10
15	7-primitive closure (v6.44)	None directly	2/10
16	Compression highest-leverage	None — opinion	1/10
17	5-prim minimal architecture	None — engineering	1/10
18	F1 complexity-class definition	Arora-Barak 2009	1/10
19	F3 framework reductions	Surveys (Buhrman-de Wolf etc.)	2/10
20	F4 6-primitive benchmark	None — experiment	2/10

Average novelty: 1.0/10. **Maximum**: 2/10 (claims 15, 19, 20). **Zero claims reach 3/10**, the threshold typically required for STOC/FOCS publication.

Part 4 — Publication Assessment

Venue	Verdict	Reason
STOC	REJECT	No new lower bound, no new separation, no new technique, no new construction.
FOCS	REJECT	Same as STOC. v6.44 7-primitive conjecture lacks formal proof.
CCC	REJECT	Misidentifies complexity classes; misses P/poly on first submission; mutual irreducibility unproven.
ICALP	REJECT	No new algorithm/complexity result/lower bound. May fit a vision track at less selective venue.
COLT	REJECT	Active learning + bandits are restatements of Dasgupta 2004, Hanneke 2007, Auer et al. 2002.
NeurIPS Theory	REJECT	Restates known ML results; no new ML-theoretic result.
JACM	REJECT	No deep result; falsification history; definition-dependent claims; unproven central conjecture.
SICOMP	REJECT	No new algorithm/complexity bound; unproven central conjecture.

Alternative Venues That Might Accept

CACM Viewpoint: A "viewpoint" article on engineering taxonomy could be published. The 7-primitive framework as an organizing principle for practitioners might pass review.

IEEE Computer / ACM Ubiquity: Practitioner-facing venues might accept the engineering taxonomy.

STOC/FOCS Workshop: A "challenges and directions" submission might be accepted as a workshop position paper, especially given the falsification history shows intellectual honesty.

These are practitioner venues, not theory venues. The submission is not theory-publishable but may be practitioner-publishable.

Part 5 — Subsuming Published Frameworks

Per the user's requirement for verdict C: the exact published frameworks that subsume every claim in v6.42-v6.44.

#	Framework	Source	Subsumes
1	Standard complexity theory	Arora-Barak 2009	All complexity-class claims (P, NP, P/poly, BPP, BQP, etc.); 10 axes subset
2	Communication complexity	Yao 1979; Kushilevitz-Nisan 1997	Communication axis (L6); information complexity
3	PCP theory	Arora et al. 1998; Dinur 2007	Verification axis (L7); V primitive
4	Algorithmic information theory	Li-Vitányi 2008	Compression axis (L2); C primitive; K(f) bound
5	Parallel complexity (NC)	Brent 1974	Scheduling axis (L4, L5); S primitive
6	Active learning + bandit theory	Dasgupta 2004; Auer et al. 2002	Adaptivity axis (L8); A primitive
7	Distributed computing	FLP 1985; Lynch 1996	FLP impossibility (CE E)
8	Quantum complexity	Bernstein-Vazirani 1997; Aaronson 2010	Quantum primitive (Q); CE B
9	Proof complexity	Cook-Reckhow 1979; Razborov 1998	Verification beyond PCP
10	Average-case complexity	Impagliazzo 1995; Spielman-Teng 2001	CE G; smoothed analysis
11	Interactive proofs	Shamir 1992; Goldwasser-Sipser 1986	IP = PSPACE (CE D)

The union of these 11 frameworks subsumes every claim in v6.42-v6.44. The submission's contribution is the organization of these frameworks under a non-standard 5/7-primitive taxonomy, not any new theorem within them.

Part 6 — Final Verdict

6.1 Assessment

· v6.42 main theorem **falsified** by v6.43's P/poly counterexample (already known in literature).

· v6.43 "engineering taxonomy" is **a reformulation** of standard axes under non-standard names.

· v6.44 7-primitive theorem is **HYPOTHESIZED** and **definition-dependent**. Components are all already known.

· No new lower bound, no new separation, no new construction, no new technique.

· Novel elements (5/7 primitive taxonomy, engineering recommendation) are definition-dependent and unsupported as theorems.

6.2 Final Verdict

C. Contains publishable synthesis but no new theorem.

The submission organizes known complexity-theory results under a non-standard 5/7-primitive taxonomy. Every claim is subsumed by 11 published frameworks (standard complexity theory, communication complexity, PCP, AIT, parallel complexity, active learning, distributed computing, quantum complexity, proof complexity, average-case complexity, interactive proofs). No genuinely new theorem exists.

The synthesis may be publishable as a position/survey paper in practitioner venues (CACM Viewpoint, IEEE Computer) but is not theory-publishable at STOC/FOCS/JACM/SICOMP.

6.3 Honest PC Acknowledgment

This PC review was conducted with maximum skepticism per the user's instructions. The submission chain (v6.42 → v6.43 → v6.44) shows intellectual honesty — the authors falsified their own v6.42 theorem in v6.43, then attempted to recover in v6.44 — but **honesty is not novelty**.

The v6.44 7-primitive framework, as an engineering taxonomy, has practitioner value. As a theorem, it is unsupported. The PC's verdict (C) reflects this distinction: the synthesis is publishable as a position/survey paper, but no genuinely new theorem exists.

If the authors wish to publish in a theory venue, they must: (1) formalize the primitive definition beyond complexity-class relabeling; (2) prove a separation that is not already known (e.g., a new BQP vs P/poly result, or a new lower bound on a specific axis); (3) drop the closure claim until a positive proof exists.

Appendix — Independent Verdicts Summary

Claim	Independent verdict
v6.42 main theorem (5-primitive closure)	INCORRECT (falsified by P/poly)
v6.42 L1 (10 axes exhaustive)	INCORRECT (6+ axes missing)
v6.42 L2-L10 (per-axis bounds)	Already known (Brent, Mehlhorn-Schmidt, PCP, etc.)
v6.42 L11 (amortization bound)	Unsupported (no citation)
v6.42 L12 (mutual irreducibility)	Incorrect (contradicted by class inclusions)
v6.43 Counterexamples A-G	Already known (standard TCS material)
v6.44 7-primitive closure	Unsupported (HYPOTHESIZED, definition-dependent)
v6.44 F1 formal definition	Definition-dependent (relabeling)
v6.44 F3 framework reductions	Mostly correct (5 corrections; no new theory)
v6.44 F4 6-prim benchmark	Empirical (engineering, not theory)

Final verdict: C. Contains publishable synthesis but no new theorem. Subsumed by 11 published frameworks.